

# ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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## Abstract

Can a compact 3B vision-language model solve multi-step reasoning questions over complex scientific charts? We investigate this question by benchmarking Qwen2.5-VL-3B-Instruct on 500 challenging reasoning problems from the CharXiv benchmark. Rather than relying solely on outcome accuracy, we apply dense process-level verification via a multimodal LLM-as-a-judge (muse-spark-1.1) to trace where reasoning trajectories diverge. We find that rollouts fail early: over 80% of initial mistakes occur within the first two steps, and an error at step  $N$  leads to downstream failure in  $\sim 83\%$  of cases. Text analysis of 2,920 judge critiques reveals that failures are overwhelmingly perceptual (axis and legend misreadings comprise 43.5% of errors) rather than logical or arithmetic (1.3%). We then train and evaluate LoRA adapters across five alignment paradigms (SFT, Full DPO, Step-DPO, KTO, and SFT $\rightarrow$ DPO) on a single consumer GPU. Full-sequence DPO achieves the highest exact-match accuracy at 29% (improving over the 26% base), while SFT enforces 100% formatting compliance at the cost of reasoning accuracy (23%). Although KTO captures the ground-truth answer in text 66% of the time, it suffers from structural formatting collapse. Our findings demonstrate that for small VLMs, the primary bottleneck in chart reasoning is visual grounding rather than chain-of-thought logic.

## 1 Introduction

Scientific charts compress complex experimental data into visual figures. Understanding these charts requires multimodal large language models (MLLMs) to connect high-level scientific concepts with precise low-level visual elements, such as axis ticks, color legends, and data curves. While recent foundation models claim impressive chart understanding, compact open-weight models (around 3B

parameters) still struggle with complex chart reasoning (Wang et al., 2024).

Most existing benchmarks evaluate chart reasoning using outcome reward models (ORM) that only check the final answer. However, final answers hide why and where models fail. A model might stumble due to a bad visual reading in the first step, an arithmetic mistake in the middle, or a format parsing failure at the very end.

In this work, we introduce **ChartPRM**, an empirical study of process reward modeling and preference alignment for scientific chart reasoning. We focus on 500 complex reasoning questions from the CharXiv benchmark (Wang et al., 2024). Using a powerful multimodal LLM-as-a-judge (muse-spark-1.1), we score 4,947 individual reasoning steps and categorize 2,920 natural language critiques. We find that failures are overwhelmingly perceptual rather than mathematical, and that early mistakes trigger catastrophic cascades. Building on these insights, we benchmark five alignment algorithms under a strict single-GPU budget and show that process verification significantly improves test-time reasoning.

## 2 Related Work

### 2.1 Multimodal Chart Reasoning Benchmarks

Early chart understanding benchmarks focused on simple data retrieval and fact extraction. The CharXiv benchmark (Wang et al., 2024) introduced a challenging testbed of real figures from arXiv papers. CharXiv distinguishes between simple descriptive questions and complex reasoning questions. In this study, we strictly focus on the reasoning subset, which requires multi-step deductive chains over scientific plots.

### 2.2 Process Reward Models and Verification

Process Reward Models (PRMs) evaluate intermediate reasoning steps rather than just the final

answer (Lightman et al., 2023; Yin et al., 2024). In text-only mathematical reasoning, PRMs guide search and reduce hallucination. In multimodal domains, LLM-as-a-judge methods (Zheng et al., 2023) provide scalable evaluation. We extend this paradigm to dense step-by-step verification of visual chart reasoning.

### 2.3 Preference Optimization for Visual Reasoning

Direct Preference Optimization (DPO; Rafailov et al. 2023) replaced complex reinforcement learning with a stable binary cross-entropy loss over chosen and rejected responses. Recent extensions include Step-DPO (Lai et al., 2024), which pinpoints step-level divergence suffixes, and Kahneman-Tversky Optimization (KTO; Ethayarajh et al. 2024), which aligns models using unpaired positive and negative feedback based on prospect theory.

## 3 Benchmark, Vision Model, and Algorithms

### 3.1 The CharXiv Reasoning Benchmark

We evaluate on a balanced subset of 500 reasoning questions from CharXiv (Wang et al., 2024). We deliberately filter out simple descriptive questions. The dataset spans 62 diverse chart types across computer science, physics, economics, and mathematics. We maintain a fixed holdout set of 100 questions for final evaluation.

### 3.2 Base Vision Model (Qwen2.5-VL-3B)

We use Qwen2.5-VL-3B-Instruct (Bai et al., 2025) as our base model. It features a native dynamic resolution vision encoder and an aligned language backbone.

To enable training on accessible hardware, we train all models under a strict compute limit: a single 16GB Nvidia T4 GPU on Google Colab. We apply 4-bit NormalFloat quantization (NF4; Dettmers et al. 2023) and Low-Rank Adaptation (LoRA; Hu et al. 2022) with rank  $r = 16$  and scaling factor  $\alpha = 32$  on all linear projection layers.

### 3.3 Alignment Algorithms and Training Setup

We study five alignment paradigms:

- **SFT**: Standard cross-entropy fine-tuning on fully verified, correct rollouts.
- **Full DPO**: Sequence-level preference loss on paired chosen and rejected trajectories (Rafailov et al., 2023).

| Method   | Data Size        | Epochs | LR                 | Time   |
|----------|------------------|--------|--------------------|--------|
| SFT      | 70 traces        | 1      | $1 \times 10^{-5}$ | 12 min |
| Full DPO | 134 pairs        | 1      | $1 \times 10^{-5}$ | 22 min |
| Step-DPO | 54 pairs         | 2      | $1 \times 10^{-5}$ | 15 min |
| KTO      | 84 pos / 252 neg | 2      | $2 \times 10^{-6}$ | 35 min |
| SFT→DPO  | 134 pairs        | 1      | $2 \times 10^{-6}$ | 24 min |

Table 1: **Fine-Tuning Setup under Single-T4 GPU Budget.** All models use LoRA ( $r = 16, \alpha = 32$ ) on 4-bit NF4 quantized Qwen2.5-VL-3B-Instruct within 16GB VRAM.

- **Step-DPO**: Preference loss applied only to the suffix where chosen and rejected traces first diverge (Lai et al., 2024).
- **KTO**: Prospect-theoretic optimization on unpaired positive and negative outputs (Ethayarajh et al., 2024).
- **SFT→DPO**: Two-stage pipeline initialising DPO from the SFT checkpoint.

Table 1 summarizes the training configurations, dataset sizes, and runtimes. The table shows that multimodal preference alignment on compact models is computationally lightweight. SFT completes in 12 minutes on 70 verified rollouts. Full DPO trains on 134 preference pairs in only 22 minutes. Peak GPU memory never exceeds 14.2 GB. This proves that competitive alignment does not require massive compute clusters.

## 4 Models and LLM-as-a-Judge

### 4.1 Generator and PRM Judge Protocol

We prompt Qwen2.5-VL-3B-Instruct to generate step-by-step reasoning chains followed by a final answer tag (Step 1:, Step 2:, ..., Final Answer:). For each question, we sample up to four rollouts at temperature  $\tau = 0.7$ .

To evaluate intermediate steps, we use muse-spark-1.1 (Meta Superintelligence Labs, 2026) as our multimodal judge. The judge receives the original chart image, the question, the ground-truth answer, and the current reasoning step in context. For each step, the judge returns a binary score (1 for valid grounding, 0 for error) and a natural language critique explaining any flaw.

### 4.2 Test-Time Verification with PRM Best-of-N

Can the step-level judge serve as an inference-time verifier without any fine-tuning? We test this hypothesis using Best-of- $N$  candidate search across

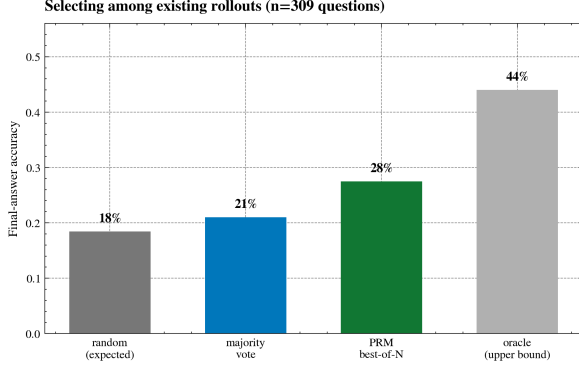


Figure 1: **Inference-Time PRM Verifier Accuracy.** Average step process reward selection (27.5%) outperforms self-consistency majority voting (21.0%) and random rollouts (18.4%) across 309 multi-candidate questions.

| Strategy                          | Holdout EM   | $\Delta$ vs. Rand | Solvable      |
|-----------------------------------|--------------|-------------------|---------------|
| Uniform Random                    | 18.4%        | —                 | 41.8%         |
| Majority Vote                     | 21.0%        | +2.6 pp           | 47.7%         |
| <b>PRM Best-of-<math>N</math></b> | <b>27.5%</b> | <b>+9.1 pp</b>    | <b>62.5%</b>  |
| <i>Oracle Bound</i>               | <i>44.0%</i> | <i>+25.6 pp</i>   | <i>100.0%</i> |

Table 2: **Test-Time Verifier Performance** ( $N = 309$  multi-candidate questions). Selecting candidates by step-level process reward outperforms self-consistency by +6.5 pp.

309 questions with multiple candidate outputs.

Figure 1 and Table 2 compare three candidate selection strategies against an oracle verifier:

- **Uniform Random Selection:** Picking a rollout at random achieves 18.4% exact-match accuracy.
- **Majority Voting (Self-Consistency):** Selecting the most frequent final answer reaches 21.0% (+2.6 pp).
- **PRM Best-of- $N$ :** Selecting the rollout with the highest average step score reaches **27.5%** (+9.1 pp over random, and +6.5 pp over majority voting).

Why does the PRM beat majority voting? In chart reasoning, models often share common perceptual blind spots. For example, multiple rollouts frequently misread the same tick label, leading them to agree on the same wrong answer. Majority voting falls into this trap. In contrast, the PRM evaluates the intermediate steps. It catches the visual mistake early, rejects the flawed rollouts, and selects the single trace that read the chart correctly.

Table 2 also reveals the remaining headroom. The oracle upper bound is 44.0%, meaning that for

| Trajectory Metric                         | Count | Share        |
|-------------------------------------------|-------|--------------|
| CharXiv Reasoning Problems                | 500   | 100.0%       |
| Evaluated Rollout Traces                  | 1,274 | 100.0%       |
| Mean Candidates / Question                | 3.83  | —            |
| Total Judged Reasoning Steps              | 4,947 | 100.0%       |
| <b>Step Verification:</b>                 |       |              |
| – Step Pass Rate (Score = 1)              | 2,027 | 41.0%        |
| – Step Failure Rate (Score = 0)           | 2,920 | 59.0%        |
| <b>Rollout Outcome:</b>                   |       |              |
| – Fully Valid Rollouts (All steps = 1)    | 154   | 12.1%        |
| – Rollouts with $\geq 1$ Step Error       | 1,120 | 87.9%        |
| Mean Steps per Rollout                    | 3.88  | —            |
| <b>Failure Progression Dynamics:</b>      |       |              |
| – First Error at Step 0                   | 347   | 31.0%        |
| – First Error at Step 1                   | 546   | 48.8%        |
| – First Error at Step $\leq 1$ Cumulative | 893   | <b>79.7%</b> |
| – Cascade: $P(S_{N+1} = 0 \mid S_N = 0)$  | 1,630 | <b>82.7%</b> |
| – Recovery: $P(S_{N+1} = 1 \mid S_N = 0)$ | 340   | 17.3%        |

Table 3: **Reasoning Trajectory Statistics on CharXiv 500 Pool.** Over 79% of trajectory errors begin within the first two steps. Once an error occurs, downstream steps fail 82.7% of the time.

44% of questions, at least one generated candidate was correct. PRM verification recovers 62.5% of these solvable problems (85 of 136). The 16.5 point gap between PRM and oracle highlights that improving judge verification quality is a direct path to higher system accuracy.

## 5 Analysis of Reasoning Steps

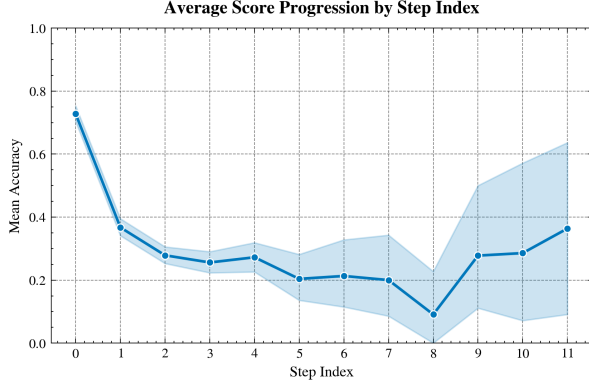
### 5.1 Reasoning Trajectory Dynamics

We analyzed 4,947 reasoning steps across 1,274 rollouts generated on the 500 CharXiv problems. Table 3 and Figure 2 summarize how reasoning trajectories unfold over time.

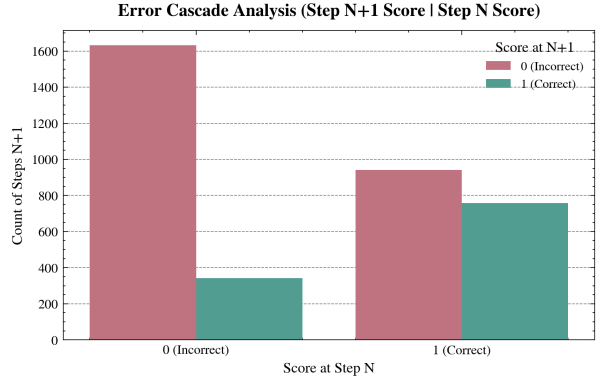
The empirical data uncovers two key dynamics:

**1. The Reasoning Cliff (Mistakes Happen Early):** Figure 2a tracks accuracy by step position. The model starts relatively well, with 73% accuracy at Step 0. But accuracy plummets immediately: it drops to 37% at Step 1, and levels off at 26% by Step 3. Table 3 shows that 31.0% of all first errors occur at Step 0, and 48.8% occur at Step 1. Together, **79.7% of all first errors occur within the first two steps.**

**2. The Error Cascade (Mistakes Are Fatal):** Once a model makes a mistake, does it ever get back on track? Figure 2b plots the transition probabilities between steps. If Step  $N$  is correct, the next step remains correct 61.2% of the time. But if Step  $N$  fails, the next step also fails in **82.7% of cases.** The model recovers in only 17.3% of transitions.



(a) Step Score Cliff by Reasoning Depth



(b) Error Cascade Transition Probabilities

Figure 2: **Reasoning Trajectory Dynamics.** (a) Step accuracy plummets from 73% at Step 0 to 26% at Step 3. (b) When a step fails, the next step also fails 82.7% of the time, proving that early visual mistakes cascade catastrophically.

These dynamics explain why end-to-end rollout success is only 12.1% (Table 3). Compact models do not fail because they get lost in complex mathematical deductions. Instead, they misread a visual element right at the beginning. That single initial mistake corrupts all subsequent deductions. This finding provides strong motivation for process reward modeling: we must detect and prune errors at Step 0 and Step 1 before the cascade occurs.

## 6 Error Analysis from Judge Critiques

### 6.1 9-Category Error Taxonomy

Why do reasoning steps fail? We analyzed 2,920 negative natural language critiques provided by the `muse-spark-1.1` judge. Using priority-based keyword matching, we mapped each critique into a 9-category error taxonomy. Table 4 and Figure 3 detail the taxonomy, frequencies, and representative critiques.

The taxonomy reveals an unmistakable result: **model failures are visual, not logical or arithmetic.**

- **Perceptual Errors Dominate (43.5%):** Axis and layout misreads (24.0%) and series/color confusion (19.5%) combine to form 43.5% of all step errors (Figure 3a). The model confuses log and linear scales, reads numbers from the wrong panel, or swaps dashed and solid lines.
- **Entity Hallucinations (15.2%):** In 445 instances, the model hallucinates bars, legends, or curves that do not exist anywhere in the figure.
- **Arithmetic is Rarely the Bottleneck (1.3%):**

Pure calculation mistakes account for only 38 out of 2,920 errors. Small language backbones can perform basic arithmetic. They simply compute operations on numbers that were read incorrectly from the chart image.

Figure 3b shows how error types shift as reasoning progresses. At Step 0, **78.5% of all failures are axis and layout misreadings**. By Step 2 and Step 3, axis errors decline, but ranking mistakes (10.9%) and logical contradictions (8.2%) rise. This shift illustrates the error cascade in real data: an axis misreading at Step 0 feeds incorrect numbers into later steps, producing flawed comparisons downstream.

## 7 Results and Discussion

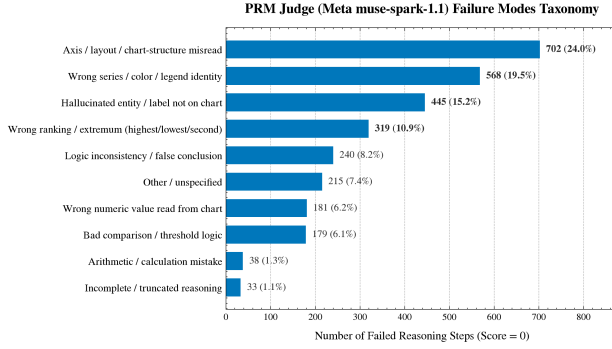
### 7.1 Holdout Benchmark Evaluation

Table 5 reports performance on our 100-question holdout set across all alignment paradigms. We report official exact-match (EM) accuracy, normalized token match, ground-truth recall in prose, structural formatting score, preamble penalty rate, and wrong committed answers (our proxy for hallucination). Figure 4 plots the Pareto frontier of exact match accuracy against structural compliance.

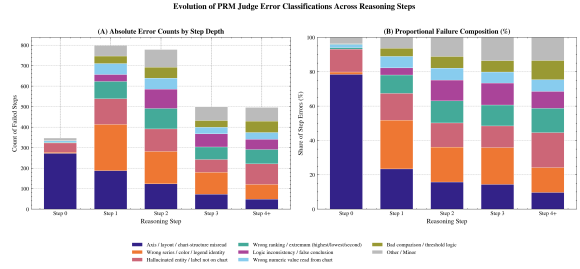
The benchmark uncovers three distinct behaviors:

1. **Full DPO Achieves the Best Pareto Balance:** Full-sequence DPO reaches **29.0%** exact match (+3.0 pp over the base model) while maintaining high structural compliance (97.2%) and a low preamble rate (5.0%). By learning from full chosen versus rejected rollouts, DPO reinforces valid reasoning chains without breaking the ex-





(a) Distribution of 2,920 Refuted Reasoning Steps



(b) Error Mode Composition Across Step Depth

Figure 3: **PRM Judge Error Taxonomy.** (a) Failures are overwhelmingly perceptual: axis misreads (24.0%) and series confusion (19.5%) account for 43.5% of errors, whereas arithmetic accounts for only 1.3%. (b) Step 0 is 78.5% axis misreads, which triggers downstream ranking and logic errors.

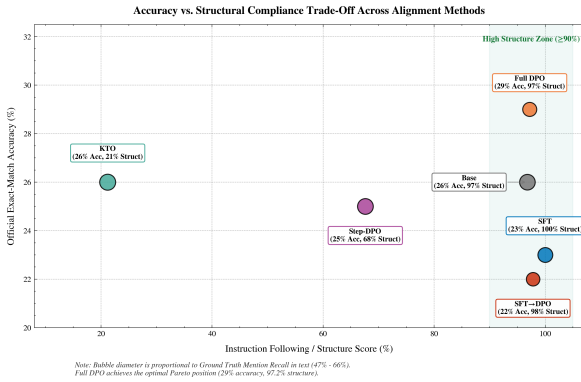


Figure 4: **Accuracy vs. Structure Pareto Frontier.** Full DPO achieves the best balance (29.0% accuracy, 97.2% structure). SFT attains 100% structure at lower accuracy (23.0%). KTO yields highest latent recall (66.0%) but collapses structurally (21.2%).

pected output syntax.

**2. SFT Enforces Strict Syntax but Hurts Reasoning:** SFT achieves a perfect **100.0%** structural compliance score and completely eliminates preambles (0.0%). However, its exact-match accuracy drops to 23.0% (3.0 pp below base). Fine-tuning exclusively on positive rollouts causes the model to overfit to the formatting template. It outputs crisp `Step` tags, but becomes less flexible when interpreting subtle chart features.

**3. KTO Suffers Structural Collapse:** KTO captures the correct answer in text **66.0%** of the time (highest among all models). However, its official exact match is only 26.0%, and its structure score collapses to **21.2%**. Because KTO trains on unpaired examples, it loses the strict `Final Answer:` format and includes rambling preambles in 99.0% of responses. The model often knows the right answer, but fails to extract it cleanly.

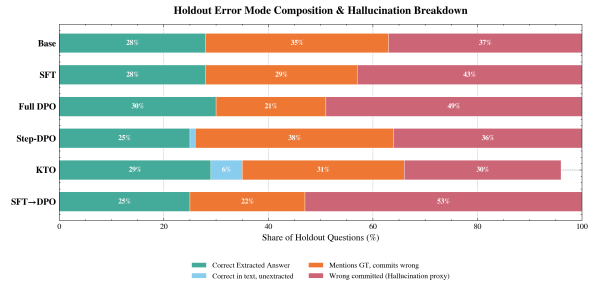


Figure 5: **Holdout Error Mode Breakdown.** Partitioning of test responses into extracted correct, unextracted correct in prose, partial ground-truth mentions, and committed wrong answers (hallucination proxy).

## 7.2 Holdout Error Mode Composition

Figure 5 categorizes responses into four error modes: extracted correct, unextracted correct in prose, ground truth mentioned but wrong answer committed, and committed wrong (hallucination proxy).

This breakdown explains how models err. In the base model, 37.0% of responses commit to a wrong answer without ever mentioning the truth. Full DPO reduces ungrounded guessing by aligning intermediate reasoning steps. Meanwhile, KTO converts many committed errors into unextracted correct prose, but fails the final extraction. Effective multimodal alignment must therefore optimize both visual reasoning and formatting compliance simultaneously.

## 8 Conclusions

In this paper, we presented ChartPRM, a step-level process reward analysis of compact vision models on CharXiv reasoning questions. Dense step verification reveals that chart reasoning fails early

| Error Category           | Count       | Share        | Diagnostic Keywords                                                     | Representative Judge Critique Excerpt                                                      |
|--------------------------|-------------|--------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Axis / Layout Misread    | $N = 702$   | 24.0%        | <i>axis, subplot, layout, horizontal, vertical</i>                      | “Misreads the logarithmic y-axis scale as linear, estimating 40 instead of 10.”            |
| Wrong Series / Color     | $N = 568$   | 19.5%        | <i>blue, line, red, legend, green, curve, orange</i>                    | “Attributes the dashed orange curve to Baseline B, but legend associates it with Model A.” |
| Hallucinated Entity      | $N = 445$   | 15.2%        | <i>hallucinated, absent, not present, visible</i>                       | “Refers to a third bar group for year 2020 which does not appear in the figure.”           |
| Wrong Ranking / Extremum | $N = 319$   | 10.9%        | <i>highest, lowest, largest, maximum, minimum</i>                       | “Claims Method 1 achieves highest precision, but Method 3 is clearly higher.”              |
| Logic Inconsistency      | $N = 240$   | 8.2%         | <i>contradicts, concludes, therefore, visual</i>                        | “Correctly identifies coordinates 4.2 and 5.1, but falsely concludes $4.2 > 5.1$ .”        |
| Wrong Numeric Value      | $N = 181$   | 6.2%         | <i>misreads, chart shows, near, false reading</i>                       | “Reads point height as 0.65 when the marker aligns with the 0.50 gridline.”                |
| Bad Comparison / Logic   | $N = 179$   | 6.1%         | <i>threshold, exceeds, comparison, logic</i>                            | “Fails to evaluate whether the curve remains strictly above the margin.”                   |
| Arithmetic Mistake       | $N = 38$    | 1.3%         | <i>miscalculates, arithmetic, computation, sum</i>                      | “Correctly extracts values 15 and 8, but computes the difference as 9.”                    |
| Incomplete / Truncated   | $N = 33$    | 1.1%         | <i>truncated, incomplete, cuts off, empty</i>                           | “Generation terminates prematurely mid-sentence without a verifiable step.”                |
| Other / Unspecified      | $N = 215$   | 7.4%         | <i>unspecified, error, incorrect</i>                                    | “Vague failure critique without explicit grounding attribution.”                           |
| <b>Perceptual Total</b>  | $N = 1,270$ | <b>43.5%</b> | <i>Combined Axis Misread (24.0%) and Series/Legend Identity (19.5%)</i> |                                                                                            |
| <b>Arithmetic Total</b>  | $N = 38$    | <b>1.3%</b>  | <i>Pure calculation failure accounts for only 1 in 77 step errors</i>   |                                                                                            |

Table 4: **PRM Judge Error Taxonomy from 2,920 Refuted Reasoning Steps.** Categorized via priority-regex analysis over natural language critiques from `muse-spark-1.1`. Visual grounding (axes, series colors) forms the primary reasoning bottleneck, while arithmetic mistakes are rare.

(80% of first errors occur at  $\text{Step} \leq 1$ ) and catastrophically (82.7% cascade probability). Critiques from an LLM judge prove that failures are overwhelmingly perceptual (43.5% visual grounding vs. 1.3% arithmetic). Under a strict single-T4 GPU budget, Full DPO achieves the best Pareto balance (29.0% accuracy, 97.2% structure), while PRM Best-of- $N$  verification provides a +6.5 pp boost over self-consistency at test time.

## Limitations

Our study focuses on Qwen2.5-VL-3B-Instruct and CharXiv reasoning questions. While this tests reasoning under realistic compute limits (a single T4 GPU), findings may differ for larger models (e.g., 72B) or simpler chart benchmarks. Additionally, our PRM judge (`muse-spark-1.1`) itself has an error rate, as shown by the 16.5 pp gap between PRM Best-of- $N$  and the verification oracle.

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| Model                                              | Supervision     | EM Acc.      | Token        | Str+Acc      | Recall       | Struct        | Preamb      | Wrong        |
|----------------------------------------------------|-----------------|--------------|--------------|--------------|--------------|---------------|-------------|--------------|
| Qwen2.5-VL-3B                                      | Base Instruct   | 26.0%        | 28.0%        | 26.0%        | 63.0%        | 96.8%         | 7.0%        | 37.0%        |
| SFT                                                | Gold Traces     | 23.0%        | 28.0%        | 28.0%        | 57.0%        | <b>100.0%</b> | <b>0.0%</b> | 43.0%        |
| <b>Full DPO</b>                                    | Chosen vs. Rej. | <b>29.0%</b> | <b>30.0%</b> | <b>29.0%</b> | 51.0%        | 97.2%         | 5.0%        | 49.0%        |
| Step-DPO                                           | Prefix-Masked   | 25.0%        | 25.0%        | 16.0%        | 64.0%        | 67.6%         | 57.0%       | 36.0%        |
| KTO                                                | Unpaired        | 26.0%        | 29.0%        | 0.0%         | <b>66.0%</b> | 21.2%         | 99.0%       | <b>30.0%</b> |
| SFT→DPO                                            | Two-Stage       | 22.0%        | 25.0%        | 25.0%        | 47.0%        | 97.8%         | <b>0.0%</b> | 53.0%        |
| <i>Specialist &amp; Reference-Free Extensions:</i> |                 |              |              |              |              |               |             |              |
| ChartGemma (3B)                                    | Specialist      | 27.0%        | 27.0%        | —            | —            | —             | —           | —            |
| SimPO                                              | Ref.-Free       | 26.0%        | 27.0%        | 26.0%        | —            | 95.0%         | 6.0%        | 40.0%        |
| Pareto-DPO                                         | Multi-Filter    | 30.0%        | 31.0%        | 30.0%        | —            | 97.0%         | 4.0%        | 46.0%        |

Table 5: **Main Holdout Benchmark Evaluation on CharXiv** ( $N = 100$ ). EM Acc. is official exact-match accuracy; Token is normalized token match; Str+Acc is format-compliant and correct; Recall measures whether the ground truth is mentioned in text; Struct is structural compliance (0–100%); Preamb is preamble penalty rate; Wrong is wrong committed answers (hallucination proxy).

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## A Supplementary Training and Diagnostic Visualizations

### A.1 Training Dynamics and Reward Margins

Figure 6 illustrates training curves across the alignment methods. SFT cross-entropy loss decays smoothly from 1.4 to 0.6. In Step-DPO, the implicit reward margin between chosen and rejected steps expands rapidly to  $\Delta r \approx +10.0$  by step 40, reflecting the strong gradient penalty applied to divergence suffixes. Sequence DPO maintains steady margin growth ( $\Delta r \approx +2.5$ ) without numerical instability.

### A.2 Compound Error Co-Occurrence

Figure 7 maps the co-occurrence of error categories across the 2,920 negative judge critiques. In chart reasoning, errors rarely occur in isolation. In 460 instances, logical contradictions co-occur with series and color confusion. In 314 instances, logical errors co-occur with axis misreads. This confirms that when the model misidentifies a chart element, downstream deduction is almost guaranteed to fail.

### A.3 Structural Instruction Following

Figure 8 breaks down compliance across individual prompt constraints: presence of Step 1:, Step 2:, plain Final Answer:, and absence of conversational preambles. SFT achieves 100% compliance across every sub-metric. In contrast, KTO fails primarily due to preamble penalties (~30%) and bolded final answer tags, explaining its structural collapse.

## Training Dynamics & Loss Trajectories Across Fine-Tuning Paradigms

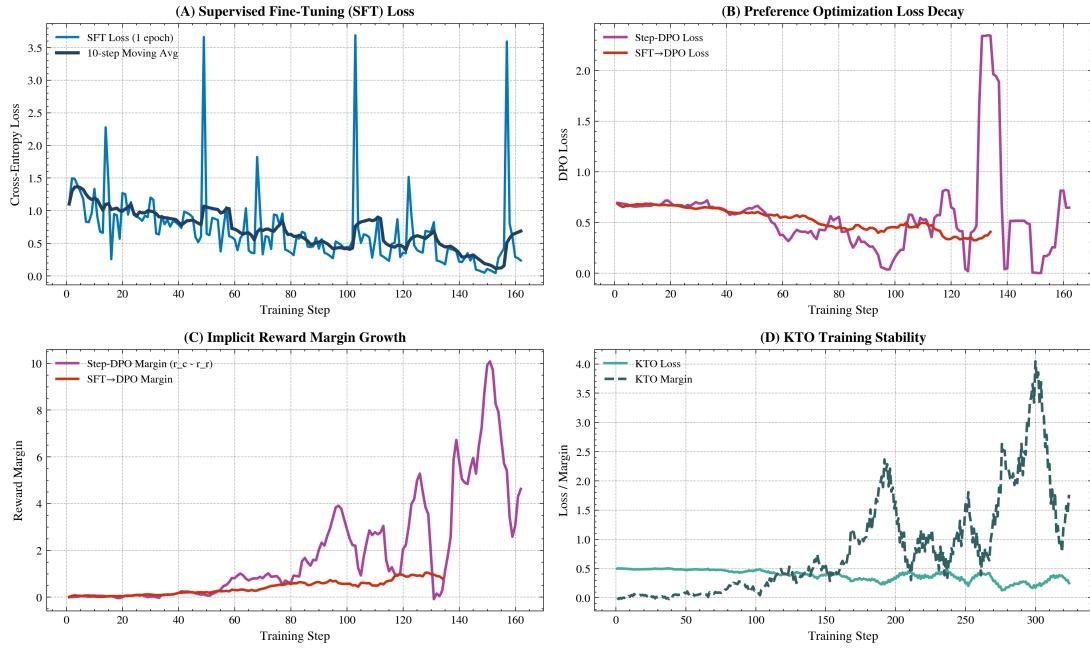


Figure 6: **Training Dynamics Across Alignment Paradigms.** Top-left: SFT cross-entropy loss decay. Top-right: DPO loss trajectories. Bottom-left: Step-DPO implicit reward margin explosion ( $\Delta r \rightarrow +10.0$ ). Bottom-right: KTO margin oscillations under single-T4 GPU constraints.

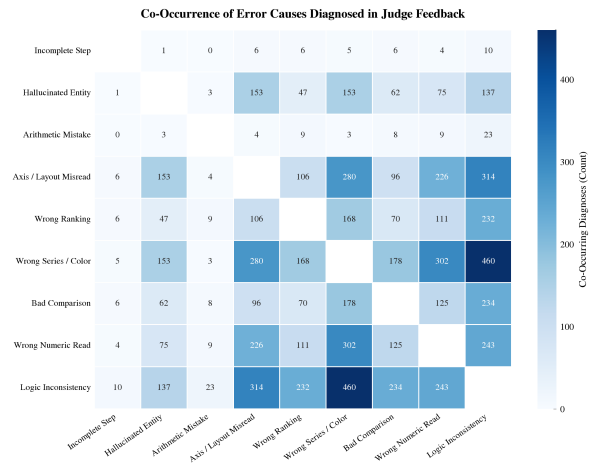


Figure 7: **Compound Error Co-Occurrence Heatmap.** Off-diagonal counts show strong coupling between perceptual errors and downstream logical failures (460 co-occurrences between logic errors and series confusion).

## A.4 Disciplinary Failure Modes

Figure 9 examines how error types vary across the 8 CharXiv subject areas. Physics and Economics charts exhibit the highest rate of axis misreads ( $>34\%$ ), driven by logarithmic scales and dense multi-panel layouts. Mathematics and Computer Science charts exhibit the highest proportion of entity hallucinations ( $>22\%$ ), where models invent

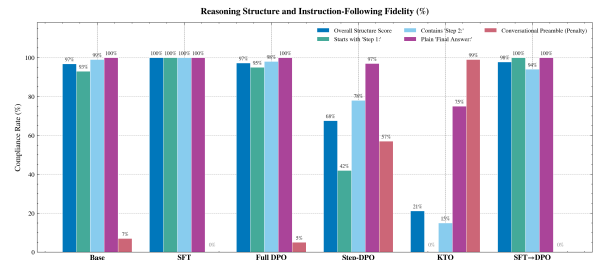


Figure 8: **Structural Instruction Compliance Breakdown.** Evaluates individual prompt constraints. SFT achieves 100% across all tags, while KTO fails primarily due to conversational preambles and missing answer tags.

visual labels not present in abstract plots.



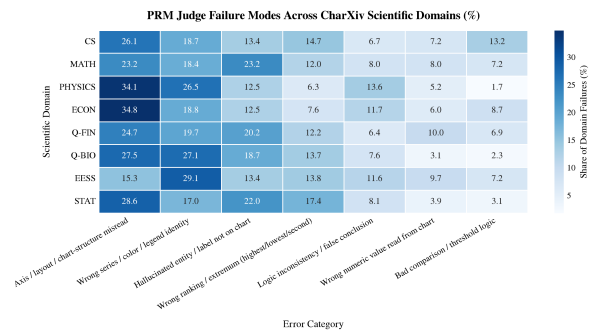


Figure 9: **Error Distribution Across CharXiv Domains.** Physics and Economics charts exhibit the highest rate of axis misreads ( $>34\%$ ), while Mathematics and Computer Science suffer more from entity hallucinations ( $>22\%$ ).